2.4 Describing the data structure API: Priority queues

Before delving into the topic of the chapter, let's take a step back.

As explained in appendix C, each data structure can be broken down into a few lower-level components:

1. An **API**. The API is the contract that a data structure (DS) makes with external clients. It includes methods definitions but also some guarantees about the methods' behavior that are provided in the DS' specification. For example, a priority queue (PQ) provides these methods and guarantees:
   a) `top()`: Returns and extract the element with the highest priority.
   b) `peek()`: Like `top` it returns the element with the highest priority, but without extracting it from the queue.
   c) `insert(e, p)`: adds a new element *e* with priority p to the PQ.
   d) `remove(e)`: removes element *e* from the queue.
   e) `update(e, p)`: changes the priority for element e and sets it to p.
2. **Invariants**: (optional) internal properties that are always hold true throughout the life of the data structure. For instance, a sorted list would have one